S26 Rate of Decay and Uses of Radionuclides | 2013 Q6

This question tested candidates' knowledge and understanding in Mechanics in the context of bungee jumping. The overall performance was only fair. In (a)(i), most candidates knew that the jumper was in free fall at the start. However, only the most able ones described correctly the change in acceleration after the cord began to stretch. Many candidates wrongly thought that the jumper began to decelerate once the cord stretched. In (a)(ii), most demonstrated an understanding that gravitational potential energy changed to kinetic energy during the fall. However, not many mentioned that all the gravitational potential energy changed to elastic potential energy in the cord at the lowest point. In (b), only the more able candidates explained clearly that the elastic cord lengthened the stopping time, hence reducing the net force acting on the jumper. In (c), candidates knew that contact area was larger for a 'full body harness' but few were able to apply the concept of pressure in their explanations.

S26 Rate of Decay and Uses of Radionuclides | 2013 Q9

In (a), candidates knew how to find the decay constant and the number of C-14 nuclei by different methods. In (c), some candidates confused N and N· or A and A·, thus wrongly substituted the number of C-14 nuclei into the formula and failing to find the correct age of the wood sample.

S26 Rate of Decay and Uses of Radionuclides | 2014 Q10

Parts (a) and (b)(i) were well answered. It revealed that most candidates had a good understanding of the properties of ·-particles and the meaning of activity. In (b)(ii), only the more able ones knew that the power is given by the product of energy per decay and the activity found in (a)(i). Some wrongly used the total number of plutonium atoms in their calculation instead of the activity. In (b), for those who considered the number of half-lives elapsed in order to find the decrease in number of active atoms or activity, many obtained the correct numerical answer.

S26 Rate of Decay and Uses of Radionuclides | 2018 Q10

Part (a) was well answered though some candidates commented that atoms with more neutrons or greater mass numbers were more stable. In (b)(i), the answers of some candidates were far from precise e.g. 'as the ionizing power is random or the type of radiation emitted is random'. Weaker candidates only focused on the EHT device. In (b)(ii), many candidates wrongly held that the Ra-226 source was not pure and thus emitted all three kinds of radiation. It seems that candidates did not fully understand the mechanism of a decay series. In (b)(iii), quite a number of them pointed out that the sparks were related to the strong ionizing power of · radiation, but many just suggested using a cloud chamber to distinguish different types of radiations emitted instead of describing a way to verify the sparks were caused by the · radiation.

S26 Rate of Decay and Uses of Radionuclides | 2021 Q9

This question tested candidates' knowledge and understanding of radioactivity in the context of radioactive potassium contained in bananas. The performance of candidates was good. In (a)(i), most candidates knew that · decay occurred. Candidates' performance was fair in (a)(ii) as some explained the answer in terms of the ionising power or range of · particles instead of penetrating power. Many candidates knew how to tackle part (b). However, quite a number of them made mistakes in unit conversions or mistook N as the number of moles of radionuclides.

Candidates did poorly in this question. Most knew the direction of the magnetic field in (a) but failed to sketch precisely the field pattern. In explaining the electromagnetic induction between the transmitting and receiving coils in (b)(i), quite a number of the candidates failed to spell out explicitly that the transmitting coil produced a changing magnetic field. In (b)(ii), very few candidates realised that the amplitude of the voltage induced in coil *R* would double when the frequency was doubled. In (c), not many candidates realised that the use of non-metallic back covers for mobile phones was to facilitate the passage of magnetic field between the transmitting and receiving coils. Some wrongly thought that 'electric shock may occur as metals are conductors' or 'current may flow through the metallic cover and cause a heating effect'.

S26 Rate of Decay and Uses of Radionuclides | 2025 Q12

This question required candidates to apply the principles of radioactive decay and their understandings of the motion of ions in a magnetic field. The overall performance was satisfactory. In (a)(i), about two thirds wrote a correct equation, although some mistakenly argued that a neutron is emitted during the decay. Around 80% found the decay constant in (a)(ii). Performance in (a)(iii) was also satisfactory, though some confused the count rate of the ancient wood sample with that of the fresh wood. About two thirds correctly stated the meaning of isotopes in (b)(i), and about half sketched the correct path in (b)(ii). The performance in (b)(iii) was poor. Many did not mention that the magnetic force is perpendicular to the velocity.